A RADICIAL MORPHISM OF ALGEBRAIC SPACES IS UNIVERSALLY INJECTIVE

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 67-70 of sources/stacks-spaces-more-morphisms/pinned-original-source.tex; proof span is lines 72-81, the complete author proof (8 body lines).

Editorial note. Navigation stubs: 1 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The definition of radicial morphisms of algebraic spaces (same chapter, immediately preceding) and the lifting characterization of universal injectivity (Morphisms of Spaces, universally-injective); cited inputs with the argument written in the retained span.

Source reference. Original lemma 040X: sources/stacks-spaces-more-morphisms/pinned-original-source.tex, source lines 81–81. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Lemma 0.1. *A radicial morphism of algebraic spaces is universally injective.*

Proof. Let S be a scheme. Let $f : X \to Y$ be a radicial morphism of algebraic spaces over S. It is clear from the definition that given a morphism $\mathrm{Spec}(K) \to Y$ there is at most one lift of this morphism to a morphism into X. Hence we conclude that f is universally injective by Morphisms of Spaces, Lemma 040X. □

Date: stacks-project:spaces-more-morphisms.tex:local-snapshot-20261011; GNU FDL 1.2 or later.